

25.6 ASSESSMENT OF EFFICIENT MARKETS HYPOTHESIS

The Efficient Markets Hypothesis (EMH) is a theory that suggests that financial markets are informationally efficient, meaning that asset prices reflect all available information at any given time. While this hypothesis has been widely accepted in the academic world, it has also received significant criticism from various scholars and practitioners in finance. Some of the main criticisms of the EMH are:

- a) **Behavioural finance:** We know that the EMH is based upon an assumption of the rationality of investors, but the studies conducted under behavioural finance argue that investors are not always rational, and their decisions are derived from their emotions, biases, or incomplete information, causing inefficiencies in the market. This means that market prices may not always reflect all available information, as suggested by the EMH.
- b) **Historical contradictions:** Some historical anomalies or contradictions cannot be explained by the EMH. For example, the momentum effect, where stocks that have performed well in the past tend to continue to perform well in the future, and this tendency contradicts the EMH which states that future prices cannot be predicted based on past prices.
- c) **Insider trading:** The existence of insider trading suggests that some market participants have access to information that is not available to the general public, indicating that the market may not be as efficient as the EMH suggests.
- d) **Market bubbles:** The EMH contends that the price of a security tends to its intrinsic value i.e., it is neither overvalued nor undervalued. But, the cases of market bubbles such as the dot-com bubble of the late 1990s and the housing bubble of the mid-2000s suggest otherwise. In both cases, asset prices soared far beyond their true intrinsic value, leading to a crash when the bubble eventually burst.
- e) **Information asymmetry:** The EMH assumes that all market players have equal information i.e., nobody has more information than the others. Do you think so? In reality, some participants may have more or better information than others. This creates an information asymmetry, which can lead to market inefficiencies.
- f) **Limited arbitrage:** The EMH argues that the arbitrage process in the market corrects these inefficiencies caused by over and under-valuation

of the stocks. However, in reality, arbitrage opportunities are often limited by factors such as transaction costs, liquidity constraints, and regulatory restrictions, which may prevent market prices from reflecting all available information.

So, the overall conclusion is that the above criticisms suggest that the EMH may not accurately describe the behaviour of financial markets and that market inefficiencies may be more common than the EMH suggests.

25.7 APPLICATIONS OF THE EMH

Here are some main applications of the EMH:

- a) **Index Funds:** The EMH suggests that it is impossible to consistently outperform the market by picking individual stocks. In other words, the market prices of the stock are already at their best values and one stock cannot perform better than the market return. Therefore, investors can choose to invest in index funds that track the performance of the overall market.
- b) **Passive Investing:** Passive investing means an investing strategy which aims to track the performance of a market index (such as Sensex or Nifty) instead of thinking that some stock will perform better than the market. The EMH supports the idea of passive investing, which involves buying and holding a diversified portfolio of stocks for the long term, rather than trying to time the market or pick individual stocks.
- c) **Rational Expectations Theory:** The EMH is closely related to the Rational Expectations Theory, which suggests that individuals make decisions based on all available information and that markets adjust rapidly to new information.
- d) **Academic Research:** The EMH has been the subject of a great deal of academic research, particularly in the fields of finance and economics. Researchers have used EMH as a framework to test various hypotheses about market behaviour.
- e) **Regulatory Policy:** The role of regulatory authorities like the Securities Exchange Board of India (SEBI) may get diluted by implication of the EMH. The EMH has implications for regulatory policy, as it suggests that attempts to regulate markets or prevent insider trading may be futile since all available information is already reflected in stock prices.
- f) **Investment Strategies:** The EMH suggests that traditional investment strategies, such as technical analysis and fundamental analysis, may be ineffective. Instead, investors may wish to focus on strategies such as low-cost index funds, diversification, and risk management.

In a nutshell, the EMH has had a significant impact on the way investors think about financial markets and has helped to shape investment strategies and regulatory policies.

25.8 APPLICABILITY OF THE EMH IN INDIA

In the Indian context, the applicability of the EMH has been a subject of much debate and research over the years. Here is a brief literature review on the same:

- Bhatia et al. (2017) conducted a study to test the weak form of the EMH in the Indian stock market using daily stock returns for the period 2006-2016. Their findings suggested that the Indian stock market is not weak-form efficient, as there is evidence of predictable patterns in stock returns that can be exploited for abnormal profits.
- Kumar and Kamaiah (2017) examined the semi-strong form of the EMH in the Indian stock market using event study methodology. They found that the Indian stock market is partially efficient in terms of reacting to news and events, but there is still room for abnormal profits to be made.
- Patro et al. (2009) analyzed the strong form of the EMH in the Indian stock market by examining insider trading activity. They found that insiders in the Indian market were able to generate abnormal profits, which suggests that the market is not strong-form efficient.
- Gupta and Sehgal (2013) investigated the efficiency of the Indian stock market in the context of the 2008 global financial crisis. They found that the market was largely efficient during the crisis period, but some anomalies suggest the presence of market inefficiencies.
- Bhattacharya and Mukherjee (2017) analyzed the efficiency of the Indian stock market during the period 2006-2016 using a combination of technical and fundamental analysis. Their findings suggested that the market is largely efficient, but there are still some anomalies that can be exploited for abnormal profits.

25.9 LET US SUM UP

The Efficient Markets Hypothesis is a financial theory that states that in an efficient market, a stock cannot perform better than the market as a whole. Three forms of efficiency can be found in the market: a) weak form, b) semi-strong form, and c) strong form. Weak form means that all past information has been absorbed by the market to arrive at the current prices, semi-strong form means that all publically available information is incorporated in the prices, and strong form means that all information (including inside information) is reflected by the market. The Capital Asset Pricing Model (CAPM) is an asset pricing model which predicts the expected return on a security. According to the model, the expected return on a security is linearly

and positively related to the systematic risk. The CAPM and the EMH are closely related. The CAPM assumes that the EMH holds good and the EMH can be tested using the CAPM. There are various methods to do this, but the Black, Jensen, Scholes (1972) is one of the most popular techniques.

25.10 KEY WORDS

The Efficient Markets Hypothesis (EMH) : The Efficient Markets Hypothesis (EMH) is a theory in finance that suggests that financial markets are “efficient,” which means that asset prices fully reflect all available information at any given time.

Weak Form of EMH : The weak form of EMH asserts that all historical price and volume data is already incorporated into current stock prices, making it impossible to profit from analyzing past trends or patterns.

Semi-Strong Form of EMH : The semi-strong form of EMH asserts that all **publicly available information** is already reflected in stock prices, making it impossible to achieve abnormal returns through fundamental analysis.

Strong Form of EMH : The strong form of EMH asserts that **all information, both public and private**, is already reflected in stock prices, making it impossible to achieve abnormal returns through any form of analysis or insider information.

Capital Asset Pricing Model (CAPM) : The CAPM defines that the expected return of an asset (or a security portfolio or stock) is linearly and positively related to its systematic risk (or undiversifiable risk).

Characteristic Line : A characteristic line is a straight line formed using regression analysis that summarizes a particular security’s systematic risk and rate of return.

25.11 SOME USEFUL BOOKS

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25.12 ANSWERS / HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) Suggests that financial markets are “efficient,” which means that asset prices fully reflect all available information at any given time. The hypothesis states that it is impossible to consistently achieve returns over average market returns because any new information that enters the market is immediately and fully reflected in the asset prices.
- 2) Weak Form, Semi-Strong Form, Strong Form
- 3) Rationality, Information is freely available, No asymmetricity of information, prices adjust quickly to new information, no individual investor can influence prices, no transaction costs.

Check Your Progress 2

- 1) The CAPM assumes that markets are efficient. The CAPM states that the expected return of an asset (or a security portfolio or stock) is linearly and positively related to its systematic risk (the part of total risk which is beyond the control of the firm or the investor; it is determined by various economic, social and political factors).
- 2) Refer to Section 25.5

UNIT 26 FINANCIAL STABILITY AND RELATED ISSUES

Structure

- 26.0 Objectives
- 26.1 Introduction
- 26.2 Concept of Financial Stability
- 26.3 Factors Affecting Financial Stability
- 26.4 Issues in Financial Stability
- 26.5 Challenges in Financial Stability
- 26.6 Risks and Financial Instability
- 26.7 Stability Measures for Ensuring Financial Stability
- 26.8 Financial Stability and Development Council
- 26.9 Financial Stability Report
- 26.10 Let Us Sum Up
- 26.11 Key Words
- 26.12 Some Useful Books
- 26.13 Answers / Hints to Check Your Progress Exercises

26.0 OBJECTIVES

After studying this unit, you will be able to:

- explain the concept of financial stability;
- identify factors affecting financial stability;
- discuss various issues in financial stability;
- state various challenges in financial stability;
- identify the major sources of financial instability; and
- suggest policy measures to attain financial stability.

26.1 INTRODUCTION

Financial stability is one of the important objectives of the Reserve Bank of India (RBI) besides macroeconomic stability. The RBI takes the key responsibility for promoting financial stability in the country and ensuring a sound financial system.

After the global financial crisis of 2008, India witnessed a deceleration in economic growth, a decline in exports, and volatility in the stock market. The

banking system too faced certain challenges such as growing NPAs. After the global financial crisis, the RBI started focusing rigorously on the financial stability of the Indian economy.

A sound financial system is the backbone of any country and so is the financial stability. In today's open economy, it is very important to have a sound financial system that can minimize the domino effect of any crisis. RBI has implemented several measures to promote financial stability during the time of crisis like strengthening the regulatory framework, improving risk management practices among financial institutions, and enhancing the supervisory and monitoring framework etc.

Several institutions and mechanisms have also been established by the RBI to promote financial stability. For instance, the Financial Stability and Development Council (FSDC) was established in 2010 to promote coordination and cooperation among the various financial regulators and other stakeholders in India.

Reserve Bank of India keeps a regular check on the financial health of the various financial components like banks, NBFCs, etc. to identify any kind of stress or risk to take timely action to avoid any adversities in the financial system and to attain the objective of financial stability and thereby economic growth.

26.2 CONCEPT OF FINANCIAL STABILITY

Financial stability refers to a financial system that is resilient to any shock or disruptions. The financial system consists of financial institutions like banks, non-banking financial institutions, financial markets, and financial instruments. Financial stability is achieved when the market volatility is low. The financial stability in a country helps ensure the efficient functioning of the entire financial system which would result in promoting economic growth and reducing poverty. Financial stability in a country ensures:

- a) sufficient availability of capital to financial institutions to ensure capital adequacy norms are met.
- b) sufficient risk management system to ensure protection against credit risk, liquidity risk, market risk etc.
- c) availability of well-developed infrastructure for trading and payment settlement.
- d) robust regulatory framework to ensure that all legal compliance is being observed and that laws and regulations are not overlooked.

26.3 FACTORS AFFECTING FINANCIAL STABILITY

Various factors have a bearing on the financial stability of the financial system of India and these factors are:

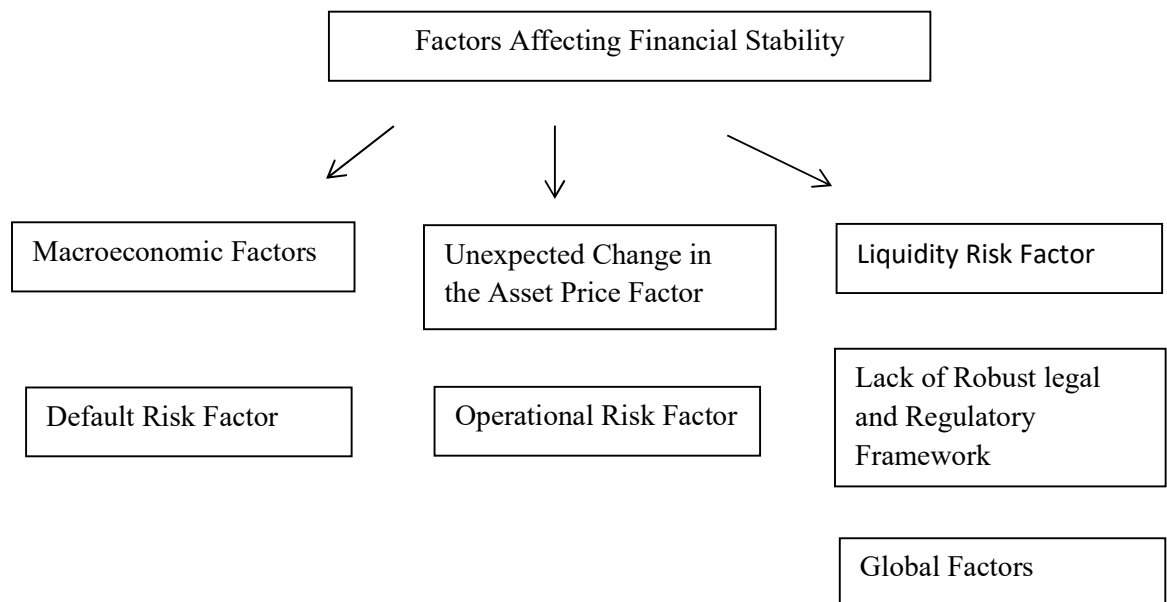


Fig. 26.1: Factors Affecting Financial Stability

- a) **Macroeconomic factors:** Factors like economic growth, interest rates, inflation, exchange rates, monetary and fiscal policy etc. have an impact on the financial stability of India. For instance, if inflation continues to grow and interest rates continue to increase, it may reduce the availability of credit to various sectors, result in high default rates, etc. and ultimately may lead to financial instability in a country.
- b) **Default risk factor:** The risk of default on loans and other credit instruments can have a significant bearing on financial stability. For instance, in case a large number of borrowers start defaulting on their loans, this may result in a liquidity crisis in the banks and will reduce the confidence of the savings holders, investors and others in the financial system of the country.
- c) **Unexpected change in asset prices factor:** The change in asset prices, like stock prices or real estate prices, have a bearing on the financial stability of the country. The rapid increase in asset prices may result in asset bubbles, which may burst at any time and can lead to financial instability in the country.
- d) **Operational risk factor:** Operational risk refers to the risk due to operational inefficiency which decreases the operational profits, sometimes results in losses and threatens the financial capabilities of any component of the financial system to meet the operational expenses or routine expenses and finally can impact the financial stability.
- e) **Liquidity risk factor:** Liquidity risk refers to the inability to meet short-term obligations in time. For instance, suppose at any point in time if the number of withdrawals from a bank suddenly increases beyond expectations then the bank may face liquidity issues which in turn increases the liquidity risk and may result in financial instability.